Navneeth Sreenivasan

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STAKE: Spatial Temporal Attention Keyframe Extractor using CNNs and CLIP Vision Language Model for Kathakali dance

Ashiq Firoz[1], Navneeth Sreenivasan[2], Nandini J. Warrier[3], and Manu Madhavan[4]

Department of Computer Science and Engineering,
Indian Institute of Information Technology Kottayam
Kerala, India
{ashiq22bcd13, navneethsreenivasan.25phd21010, nandinijw, manum}@iiitkottayam.ac.in

Abstract Kathakali is a dance form from the Indian state of Kerala where different hand gestures and facial expressions can be combined with body positions and angles to represent almost a thousand words. Translating the words from videos of Kathakali dance postures requires keyframes consisting of still postures, which in turn requires means of keyframe identification. In this work, we propose a novel approach for keyframe identification and extraction, called STAKE (Spatial Temporal Attention Keyframe Extractor), created using CNN models, attention and LSTMs. We collect videos for twenty words used in Kathakali and manually identify the keyframes from these videos, which serve as Ground Truth. We then use different CNN models such as ResNet50, DesneNet121, EfficientNetB0 and MobileNetV2 to extract the low-level features, combine them with the objects detected using the CLIP Vision Language model, and extract the temporal dependencies between image frames using Bi-LSTM. The detected keyframes, on comparison with the Ground Truth, gives 75% accuracy, and comparable precision, recall, F-1 score and Mean Absolute Error. We further plan to improve this work by balancing the dataset, reducing the model training time and conducting ablation studies to evaluate the performance of each component.

Keywords: Kathakali · Keyframes · Keyframe Extractor · CNN · CLIP · LSTM

1 Introduction

Kathakali, a classical dance form originating from the southern Indian state of Kerala, is renowned for its intricate body movements and immersive facial expressions, which serve as a medium to narrate stories drawn primarily from the folklore of Kerala. Central to Kathakali are its expressive elements - *Mudras* and *Navarasas* - that create an enticing experience for the audience. *Mudras* represent the hand gestures shown in Kathakali, whereas *Navarasas* refer to the

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facial expressions. Mainly, 24 different *Mudras* and 9 different *Navarasas* exist in Kathakali, which can be combined with different body positions and body angles, to represent almost a thousand words [1]. Due to this reason, Kathakali is also considered as a sign language. Short video clips of Kathakali performances are usually used to depict a word represented by the postures in that video clip.

The dance postures in Kathakali can be classified into still postures and transition postures. Still postures are the postures where there is no movement, whereas transition postures have some form of movement. Translating the words from videos of Kathakali dance postures requires understanding the still postures, which brings about the need for keyframes. Every video consists of image frames, which can be classified into Keyframes and Transition frames. Keyframes represent the still postures in a video, whereas transition frames are used to represent the transition postures. Kathakali dance consists of keyframes and transition frames which come one after the other.

Generally, the transition frames are blurry and do not contribute a lot in learning, and therefore, a sequence consisting of only keyframes can be used for translation. Using keyframes rather than all the frames can also help in significantly reducing the computational load, training time, which leads to increase in efficiency. Apart from sign language translation, keyframe identification is also used in other domains such as surveillance, cancer detection, polyp identification etc. [2] [3] [4]. Furthermore, Keyframe analysis can help in various tasks such as Video Summarization or Video Classification [5] [6]. Due to these reasons, keyframe identification and extraction is an area where a lot of research is going on.

In this work, a novel keyframe extraction model called STAKE (Spatial Temporal Attention Keyframe Extrctor), which combines CNN models, Vision language models as well as LSTMs, is proposed. We first collect videos of twenty words used in Kathakali and manually identify the keyframes from them, which serve as Ground Truth. We then use different CNN models such as ResNet50, MobileNetV2, EfficientNetB0 and DenseNet121 to collect low-level features of each image frame. We also use the CLIP Vision Language model to identify objects, such as *Mudras* or *Navarasas* from the image frames. The features identified by the CNN models as well as the vision language model are combined together and a Bidirectional LSTM is used to identify the temporal information between the frames. We then classify the frames into keyframes and transition frames. Accuracy, Precision, Recall, F-1 score and Mean Absolute Error are calculated by comparing the output keyframes of the model with the Ground Truth.

The rest of this paper is organized as follows. Section II comprises the Literature Survey which reports the previous methods used to perform Keyframe extraction; Section III presents the Methodology of our work in detail; Section IV reports the Experiments we ran and the results we got; and Section V concludes the work.

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2 Literature Review

Previous works focusing on sign language translation have mainly used various kinds of deep learning models and image processing techniques. Donepudi, Swapna, et al. [7] used a system comprising of YOLOv11 and LSTMs to detect the hand postures using wearable gloves and identify temporal information across the videos. Ingoley, Shilpa, and Jagdish Bakal. [8] used VGG16 and MobileNetV2 rather than LSTMs to translate Indian Sign Language postures representing numbers from zero to nine, as well as thousand, lakh (hundred thousand) and crore (ten million). Renjith, S., Manazhy Rashmi, and Sumi Suresh [9] created an architecture, termed as Time-distributed CNNs, rather than LSTMs for translation of Chinese Sign Language (CSL) and Arabic Sign language (ArSL). However, all these works have used all the frames, rather than just focusing on the keyframes, leading to longer computational time and a larger computational load, which in turn leads to a reduction in efficiency.

However, some previous works have focused on keyframe identification and extraction. Most of them have focused on deep learning and computer vision based techniques, some have used clustering-based approaches and a few have used other approaches. Some of them are mentioned below.

2.1 Deep Learning and Computer Vision based approaches

Huang, Ruobing, et al. [3] manually annotated breast cancer images to serve as Ground Truth, and developed a reinforcement learning model to compare the model output with the Ground Truth images, for use in classification. A Bi-LSTM model was used to classify the videos to detect cancer. Prakash, M. Madhura, and G. N. Krishnamurthy [4] developed a model combining MobileNetV2, UNet and attention from Transformer architecture to get the depth estimation from image frames of polyp in human intestines, and using a combination of information entropy, key points and image moment from these depth estimated image frames, identified the key frames. Mo, Clinton, et al. [10] created skeletal images of people from the images, use it as a graph and find similar skeletal images using a reinforcement-learning model which rewards or penalises the model. Zarif, Sameh, et al. [11] first removed the blurry frames using a combination of intensity, contrast, and motion detection between pixels. They then extracted the low-level features using ConvNext architecture and clustered these features using a combination of Bayesian networks, PCA and K-Medoids clustering, to get the keyframes. Zheng, Bowen, et al [12] derive control signals from the root trajectory paths of humans and developed equations to measure the difference between consecutive frames. They create a diffusion model to automatically generate keyframes and further create a synthetic dataset to test their model.

Tan, Guanghua, and Jinxi Zou [13] used an approach by calculating the colour histogram and similarity matrix of each frame, classified the videos to smaller video shots and reported the centre frame of each shot was as the keyframe. Kavitha, J., et al. [14] first removed the blurry frames manually,

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then applied a three-stage method to detect keyframes. They first applied a Multi-Criteria Decision Making system based on visual quality and motion details, then performed low-level feature extraction, normalization, weight assigning, and score calculation on top of the system, and finally, applied a RNN model with LSTM to extract the keyframes. Ashiquzzaman, Akm, et al. [15] used a combination of Bhattacharyya distance as well as Histogram of images to develop equations and created a threshold to detect keyframes, then used it to classify surveillance videos with fire and smoke. Singh, Pratibha, and Alok Kumar Singh Kushwaha [16] extracted all the frames, used a sliding window to filter out frames, and then used SURF, GIST and DNS to extract the keyframes. They measured the temporal distance between simultaneous frames using a threshold value.

2.2 Clustering-based approaches

Tan, Kailong, et al. [4] used an adaptive clustering based method similar to K-means, by initially creating random clusters, found the cluster centres, and detected keyframes from cluster centres using distance-based metrics. Tang, Hao, et al. [5] focused on extracting features in frames using ResNet, calculated the distance between the features, and used an algorithm to create clusters from the features. They extract the top 1-2% neighbours of the cluster centres as the keyframes and further perform video classification using LSTMs on the keyframes. Lv, Chaohui, and Yiyang Huang [17] removed the blurry frames using motion blur detection and used nearest neighbour clustering algorithm to find keyframes from the cluster based on similarity between the frames. Xiao, Yang Dong, Cheng Xin Min, and Ran Mao Ke [18] improved the speed of clustering by adding a lightweight-SAM model. The lightweight-SAM model is able to identify objects in the image, use these identified objects to identify the cluster centres, and use the centres for keyframe identification. Ai, Yu, Ce Xu, and Yansong Bai [19] encoded the frame bit number of each frame and then found the motion vector and coding unit depths of each frame bit. These are used to create clusters and the frame closest to the cluster centre is identified as the keyframe.

2.3 Other approaches

Rajan, Manjusha, and Latha Parameswaran [2] create a method to classify keyframes based on Genetic Algorithm. They identify the keyframe as the fittest chromosome and perform crossover and mutation across all the frames to identify the keyframes. Navyasri, M., and G. Jaya Suma [20] use cosine similarity and structural similarity between the neighbouring frames to detect keyframes. They also create a deep learning model with channel attention to classify the keyframes of sign language videos. Bhuyan, Himadri, et al. [21] first perform uniform sampling across Bharatanatyam dance videos to reduce the number of images, then classify a frame based on three different classification techniques -

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threshold value, majority voting mechanism and using Machine learning, specifically, SVM and CNN. Manually annotated images are used as the Ground Truth for comparison with the output of machine learning approach. Peer Mohamed Appa, M. A. Y., et al. [22] have used a statistical feature extraction technique, by measuring the mean and standard deviation between consecutive images and find the similarity between consecutive frames. They frames with more distance classified as keyframes and is used to identify abnormal vehicles from surveillance videos. Qiao, Shunli, and Honglin Liu. [23] identified keyframes by splitting large surveillance videos into smaller video shots, and extracting the first and last frame from these shots. These frames from the shots are combined to get the keyframes for a large video, and these are then passed through a modified Inception architecture to learn temporal dependencies to classify criminal activities.

2.4 Limitations of the existing approaches

The above-mentioned authors have used a wide variety of approaches for keyframe extraction. However, a few limitations exist across the existing approaches. The outputs of almost all the previous works, including clustering-based approaches, computer vision-based approaches, uniform sampling etc. consists of transition frames, thereby leading to inaccuracy. Specialised data such as sign language or dance data may need object detection algorithms, and using just CNNs and LSTMs could lead to lower accuracy. Reinforcement learning requires the use of designing effective reward and penalty functions, however they may be specific to different applications.

To overcome the limitations of the existing approaches, we propose a novel keyframe identification and extraction method, called STAKE. Rather than using CNN or LSTMs by themselves, we combine CNNs and LSTMs with CLIP Vision Language model to detect the objects in a video. Using a combination of the features identified from the CNN models as well as the CLIP model, along with the temporal information learnt by the LSTMs will help in the classifying frames into keyframes and transition frames.

3 Methodology

We first collected the dataset, which consists of short video clips (5-10 seconds long) of Kathakali dance, each representing a word. Next task was to extract all the frames from each of the videos following which, we manually identified the keyframes. These manually identified keyframes are considered our Ground Truth values and will be used to compare with the output of our model. After this, all the image frames, are passed to the STAKE (Spatial Temporal Attention Keyframe Extractor) model. These frames are passed through the CNN models to extract the low-level features, and are also passed through the CLIP vision language model to detect objects. These are then passed through a Bi-LSTM model to identify the temporal dependencies between the frames. Finally, these

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frames are classified between keyframes and transition frames and accuracy is calculated by comparing the output keyframes give by the model with the manually identified keyframes (Ground Truth).

3.1 Dataset collection

We used twenty Malayalam words commonly expressed in kathakali. These words, written in Malayalam , along with their English transliterations, and English meanings are provided in Table 1. Ambiguities are present while representing a word in Kathakali language. One such example is the word, *Nee* (English meaning "You"), which has two different types of representations, using two different Navarasas (facial expressions), Shringaaram and Roudram. We collected these separately, and this brought the total to about twenty one different representations from the twenty words. We recorded the video for each representation ten times with the help of an expert in Kathakali dance, with small changes in the body angles and camera movement. The data was collected in 1080p as well as 4K resolutions with some videos recorded in 30 fps and some in 60 fps. All the videos were recorded in an iPhone 17. These settings helped us ensure that the videos emulated real-world data with different resolutions, varying speed as well as from different angles. The videos were named according to the word they represented and the number of times they were performed. For example, if the word being represented was *Aana* (Elephant) and the second iteration of this word was being recorded, that video was named as *Aana_2*.

3.2 Manual keyframe identification

Our next task was to extract all the frames from the videos and name them. We followed the naming convention <word>_<iteration>_<frame_number> to name the frames. For instance, if the video currently being considered was *Aana_1*, then the frames were named as *Aana_1_001*, *Aana_1_002*, etc. Our next task was to manually identify the keyframes from all the frames in all the videos. We found that most of the words have three still positions - in the starting, middle, and the end. Therefore, we decided to select three sets of keyframes corresponding to these three still positions from each video to ensure consistency. For instance, if the three still positions from the video *Aana_1* correspond to the frame numbers 1-10, 99-104, and 185-205, then the frames *Aana_1_001* to *Aana_1_010*, *Aana_1_099* to *Aana_1_104* and *Aana_1_185* to *Aana_1_205* are extracted. These manually extracted keyframes are used as the Ground Truth values to measure the accuracy later. Some of the manually identified keyframes of the video *Aana_5* are shown in Figure 1.

3.3 STAKE Model Architecture

Our STAKE model consists of four components - the CNN models (consisting of ResNet50, DenseNet121, EfficientNetB0 and MobileNetV2), the CLIP Vision

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Malayalam words	**English transliterations**	**English meanings**
ആകാശം	*Aakasham*	Sky
ആന	*Aana*	Elephant
അച്ഛൻ	*Acchan*	Father
ഭൂമി	*Bhoomi*	Earth
ബ്രഹ്മാവ്	*Brahmaavu*	God Brahma in Hindu mythology
ചന്ദ്രൻ	*Chandran*	Moon
ഇരിക്കുക	*Irikkuka*	To sit
കാട്	*Kaadu*	Forest
നടക്കുക	*Nadakkuka*	To walk
നീ	*Nee*	You
പാദം	*Paadam*	Feet
സമുദ്രം	*Samudram*	Ocean
സത്യം	*Satyam*	Truth
ശിവൻ	*Shivan*	God Siva in Hindu mythology
സിംഹം	*Simham*	Lion
സൂര്യൻ	*Sooryan*	Sun
സ്വർഗം	*Svargam*	Heaven
വീട്	*Veedu*	House
വിഷ്ണു	*Vishnu*	God Vishnu in Hindu mythology
വൃക്ഷം	*Vriksham*	Tree

Table 1. Words used

Language model, the Bi-LSTM model, and the Classifier. The detailed STAKE model architecture is shown in Figure 2. All the frames of every video are trained by each of the CNN models and the CLIP model to get the low-level features, then these frames are trained by the CLIP Vision Language Model for object detection and the features identified by both CNN models and CLIP are combined together. These image frames are then trained by Bi-LSTM to extract the temporal dependencies between the frames. Finally, they are classified into keyframes and transition frames. Since our videos consist of both spatial features such as colours and edges, as well as Kathakali-specific semantic features such



Figure 1. Examples of manually identified keyframes

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as *Mudras* and *Navarasas*, using either CNN models or CLIP model will prove insufficient. Therefore, we decided to use both CNN models as well as CLIP Model for feature extraction.

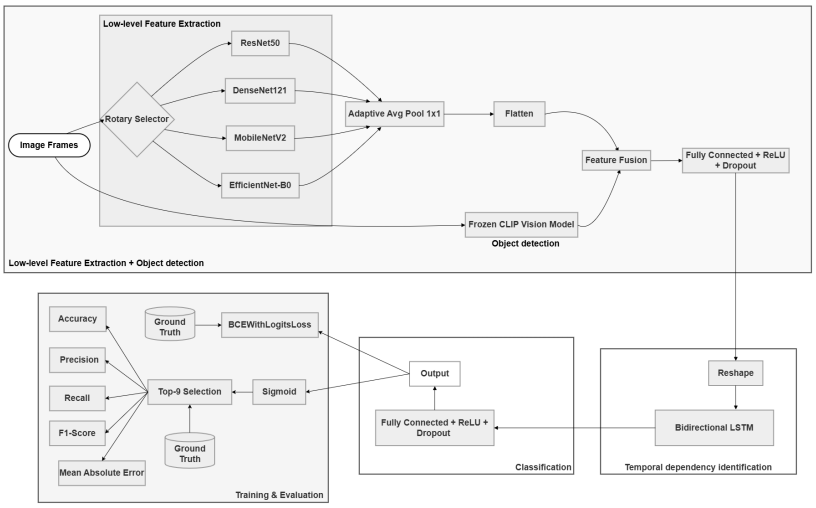


Figure 2. Model Architecture

Low-level feature extraction using CNNs All the image frames are trained separately by the various CNN models to learn low-level features such as colour, shape etc. Using a CNN model will help us in understanding the spatial information such as colours and edges across the frames. The different models which we used are ResNet50, DenseNet121, EfficientNetB0 and MobileNetV2, which helps in extracting the spatial features from the image frames.

- ResNet50 [24] is a widely used benchmark CNN architecture, consisting of skip connections, which helps gradients flow through deeper networks. It is widely used in general image-related tasks and has around 25.6 Million parameters.
- DenseNet121 [25] is a CNN model with dense layers, where each neuron in a layer is connected to each neuron in the preceding and subsequent layers. It helps in reusing features through the dense layers, and has a size of around 8 Million parameters, making it less heavy than ResNet50.
- EfficientNetB0[26] is a lightweight CNN model which consists of Squeeze-and-Excitation blocks and inverted residual blocks. It uses compound scaling to uniformly scale depth, width and resolution optimally, and has 5.3 million parameters.

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– MobileNetV2 [27] is a lightweight CNN model which utilizes depth-wise separable convolutions and inverted residuals. It has 3.5 Million parameters, making it a powerful model across edge devices.

These models are diverse with differing number of parameters, diverse architecture, and different use cases. ResNet50, though heavy on parameters, is a benchmark model and is found in almost all image-related tasks. DenseNet121 has a very complex architecture with a large number of dense layers, but this also makes it efficient for parameter reuse. EfficientNetB0 and MobileNetv2 are both lightweight models, and therefore, can be deployed in edge devices. Due to this reason, each of these models are needed for extracting the low-level features.

Each CNN model gives outputs of different sizes, which could lead to inconsistency in subsequent components. To mitigate this issue, we added an Adaptive Average Pooling layer of size 1x1 after the outputs of each CNN. These were then flattened to help ease in fusing with the outputs of the CLIP model.

High-level object detection using CLIP Vision Language model Large language models, also known as LLMs, are very large deep learning models that are pre-trained on vast amounts of text data [28]. The underlying transformer is a set of neural networks that consist of an encoder and a decoder with self-attention capabilities. The encoder and decoder extract meanings from a sequence of text and understand the relationships between words and phrases in it. The self-attention mechanism helps to learn the importance of one word with another, with multi-head attention doing it in parallel to capture various types of relationships. A vision language model is an AI system built for emulating LLMs in computer vision-related tasks [29]. VLMs can process and provide advanced understanding of video, image, and text inputs to generate text responses. CLIP (Contrastive Language-Image Pre-training) [30] is a multimodal vision language architecture which is widely used in image segmentation and object detection. Unlike other vision language models, CLIP was trained on 400 Million image-text pairs and can perform classification on datasets it was not trained on, helping it to easily capture high-level semantics (images, text etc.) [31].

To identify the Kathakali-specific entities in an image, such as various *Mudras* shown in the hand, or various *Navarasas* expressed in the face, we use the CLIP model for object detection. In our work, we use the pretrained CLIP base model, which consists of twelve layers. Instead of training the CLIP from scratch, we froze all the layers of the CLIP model and directly trained our images with CLIP. This ensures that CLIP does not overfit on our dataset, and also reduces the computational load.

Fusing the features The layers of CLIP model acts as extra layers on top of the CNN models, thereby ensuring that our architecture learns both the low-level features, such as colour and shape, as well as semantic features such as *Mudras* and *Navarasas*. In order to do that, we fused the features identified by

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each of the CNN models with the features identified by the CLIP model. This ensures us that we have four sets of feature combinations, which are listed below -

1. ResNet50 + CLIP
2. MobileNetV2 + CLIP
3. EfficientNetB0 + CLIP
4. DenseNet121 + CLIP

Three layers follow the feature combination -

1. Fully connected layer - The fused features are then connected to a fully connected layer of size 512 neurons. Each neuron in the previous layer is connected to each neuron in the next layer. The size, 512 neurons, was chosen to correspond to the size of the Bi-LSTM model
2. ReLU Activation - Activation functions are used to bring non-linearity to the model, and ReLU is one of the most common and most powerful activation functions. The equation of ReLU is shown below:

$$f(x) = \max(0, x) \tag{1}$$

3. Dropout - Dropout layers randomly turns off certain neurons across the layer, which will prevent overly relying on certain neurons and force the network to learn more robust features. A dropout of size 0.3 indicates that 30% of the neurons randomly in that layer are turned off. 0.3 is one of the most commonly used values for dropout, since it is a convenient middle point between 0 and 0.5 (choosing to drop no neurons or 50% neurons)

Addition of Temporal information using Bi-LSTM After combining the features, a Bi-LSTM model is used to train the image frames to ensure that the temporal dependencies between the images are learnt by the model. A Bidirectional LSTM (Bi-LSTM) is used to manage the vanishing gradient problem faced by RNNs and capture temporal dependencies in sequential data by monitoring information flow in forward as well as backward direction simultaneously [32]. Bi-LSTM is more efficient than a regular LSTM since it has more accuracy and consists if more connections, thereby making the model learn more information. Our Bi-LSTM consists of two layers, each consisting of 512 hidden units. The four sets of models are now as follows -

1. ResNet50 + CLIP + Bi-LSTM
2. MobileNetV2 + CLIP + Bi-LSTM
3. EfficientNetB0 + CLIP + Bi-LSTM
4. DenseNet121 + CLIP + Bi-LSTM

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Classification Head We now add the following layers -

1. Fully connected layer - A Fully connected linear layer of size 1024 is added. This ensures that both the layers of LSTM, each containing 512 neurons is flattened
2. ReLU - A ReLU Activation function is added to bring non-linearity to the model
3. Dropout- Another dropout layer of 0.3 is added, making 30% of neurons turn off at random
4. Output - This is the layer which gives the final output from the classifier. Logits layer is used as the output layer to get raw, un-normalized scores

4 Experiments and Results

4.1 Experimental setup

We used 70:10:20 train-validation-test split to evaluate our model and ran it for 5 epochs. We experimented on the number of epochs and found that 5 was the most suitable, since almost all the time, there was no large difference in the evaluation metrics after 5 epochs. Furthermore, we also observed that sometimes the models started overfitting after 5 epochs. The model training and testing was done using Google Colab's T4 GPU using Python language. The batch size was set to 1 due to computational constraints. We chose a cut-off value of 5 for the Bi-LSTM, thereby making the Bi-LSTM store information of 5 frames at a given time, leading to lesser computational overhead. The model is trained to predict either 0 (transition frame) or 1 (keyframe).

We used Sigmoid activation function to produce output probabilities and Binary Cross Entropy (BCE) to compute the loss. BCEWithLogitsLoss implementation in PyTorch, which merges these two was used for this purpose. The model trains by learning from the Ground Truth keyframes and tries to increase perform better across the epochs. The loss values are backpropogated and the weights and biases are adjusted by the model. Apart from this, we used another Sigmoid activation function at the output, after training, to interpret the predictions. The equation of sigmoid activation is shown below:

$$\sigma(x) = \frac{1}{1 + e^{-x}} \tag{2}$$

The accuracy, precision, recall and mean absolute error are calculated by comparing the output of the model with the manually identified keyframes (Ground Truth). Our Ground Truth keyframes consists of three sets of keyframes from each video, and we decided to select three keyframes from each set. These keyframes are selected based on equal distance from each other, thereby totalling to nine keyframes per video. To ensure consistency in accuracy, we select nine frames which the model detected as keyframes, having the highest confidence score predicted by the sigmoid activation function at the output, per video. This strategy helps us in reducing computational overhead by not comparing

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each and every output frame with each and every Ground Truth frame, since both are significantly high in number. These top nine keyframes are used in evaluating our metrics.

4.2 Results

The average accuracy, precision, recall, F-1 score as well as Mean Absolute Error from each of the four sets of models is shown in Table 2. The predicted image frames for the word *Aana* (Elephant) from the video *Aana_1* are shown in Figure 3.

Model	Average Accuracy	Average Precision	Average Recall	Average F-1 score	Average MAE
Model using ResNet50	0.7584	0.3333	0.4315	0.3761	0.2800
Model using DenseNet121	0.7487	0.3111	0.4027	0.3510	0.2826
Model using EfficientNetB0	0.7445	0.3016	0.3904	0.3403	0.2999
Model using MobileNetV2	0.7431	0.2984	0.3863	0.3367	0.3029

Table 2. Results of classification



Figure 3. Predicted keyframes

4.3 Observations and Analysis

All the four sets of model combinations show relatively same values for accuracy, precision, recall, F-1 score and Mean Absolute Error, despite utilizing CNN models with varying architecture, number of parameters and size. This shows that the CNN model's contribution is secondary to that of the CLIP model and the Bi-LSTM temporal modelling. However, the CNN models are still necessary to learn low-level features such as colour, edge etc. and without the addition of the CNN models, these low-level features may not be learnt by the model.

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Using CLIP model alone may be better in Zero-shot scenarios, but it could also be influenced by bias in training data. Therefore, even though the performance is dominated by the CLIP and Bi-LSTM models, CNN models are still required and removing them could lead to lower accuracy, precision, recall, F-1 score and higher MAE.

Another observation we can make is that metrics of Lightweight models like MobileNetV2 (with 3.5M parameters), EfficientNetB0 (with 5.3M parameters) and DenseNet121 (with 8M parameters) have almost the same value compared to that of heavy architectures such as ResNet50 (with 23M parameters). We can see that that increasing the number of parameters do not show a better accuracy in the proposed architecture, which leads us to conclude that lightweight CNN models are more suitable than heavier models for keyframe extraction, especially on edge devices. However, ResNet50 is a widely used benchmark model is found across almost all image-related tasks. Therefore, even though it has comparable accuracy with the other models and has a heavier architecture, usage of ResNet50 is significant for comparison.

A third observation we can make is that the accuracy is higher while the Precision, Recall and F-1 score are lower. This can be explained by the fact that our video has a significantly higher number of transition frames (1) as compared to keyframes (0). Since the keyframes are under-represented, our model trains on far more transition frames than keyframes, which could lead to a lower value of True Positives and higher values of True Negative as well as False Positive. This in turn, could lead to lower precision and recall, and therefore, a lower F-1 score, as evidenced by our results.

5 Conclusion and Future Work

Keyframe identification is one of the most important works used in various video-related tasks, such as video summarization or classification. In this work, we present a novel keyframe identification and extraction method using CNN models, VLMs and LSTMs. CNN Models such as ResNet50, MobileNetV2, EfficientNetB0 and DenseNet121 are used to identify the low-level features across videos, whereas the CLIP Vision Language model is used to identify the *Mudras* and the *Navarasas*. Bi-LSTM is used to extract the temporal dependencies between the frames and a classifier is used for frame classification. The accuracy, precision, recall, and MAE were calculated by comparing with the Ground Truth values.

This method presents a novel approach using a combination of CNN models, VLMs and Bi-LSTMs and has a wide variety of applications. However, a limitation in this work is that the usage of VLMs require more computational resources, leading to slower time for model training. Additionally, our dataset has class imbalance which affects the precision and recall, thereby affecting the F-1 score. In the future, this work can be expanded to other sign languages such as Indian Sign Language (ISL), American Sign Language (ASL) etc. We are also planning to handle the class imbalance and improve the model training time by experimenting with other models. Furthermore, we are also planning to conduct

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an ablation study by removing or fine-tuning certain components and observing how these changes affect the model.

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